Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples

Parts of an Expression

$7x + 3 - 5x + 8 + 4x - 1$

Variable Terms
(Terms WITH a variable)
$7x, -5x, 4x$

Constant Terms
(Terms WITHOUT a variable)
$3, 8, -1$

Coefficients
(Number NEXT TO a variable)
$7, -5, 4$

Combining Like Terms

You can simplify an algebraic expression by **combining like terms.** This means to combine common variable terms and contant terms.

Example: Simplify the expression below:

$7x + 3 - 5x + 8 + 4x - 1 = 6x + 10$

Examples

Directions: Simplify each expression.

1. $3x + 6x$ $9x$	**2.** $9a - 10a$ $-a$
3. $7m - 5m - 6$ $2m - 6$	**4.** $9 - 6w + 5$ $-6w + 14$
5. $7p - 1 - 9p + 5$ $-2p + 4$	**6.** $5h - 6 - 8 + 7h$ $12h - 14$
7. $5x + 7y - x + 2y + 3x$ $7x + 9y$	**8.** $26a + 2b - 4a + 10b$ $22a + 12b$

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with Exponents

Terms with the same exponents or combination of variables and exponents are considered like terms. When combined, be sure to NOT change their exponents!

Examples

9. $5x^2 + 7x^2$

$12x^2$

10. $-15n^5 + 4n^5$

$-11n^5$

11. $8c^3 - 7c^3 + 16c^3$

$17c^3$

12. $8k^2 - k - 5k + 7 - 2k^2$

$6k^2 - 6k + 7$

13. $-4a^2b + 7 + 10a^2b$

$6a^2b + 7$

14. $x^2 + 2xy - 7xy + 4y^2$

$x^2 - 5xy + 4y^2$

15. $a^2 + 7ab - 10ab + b^2$

$a^2 - 3ab + b^2$

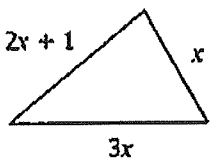
16. $4x^2 - 7 + 2x - 3x^2 + 8x - 6$

$x^2 + 10x - 13$

Geometric Applications

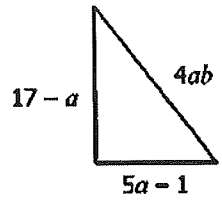
Directions: Give the perimeter of each figure as a simplified expression.

17.



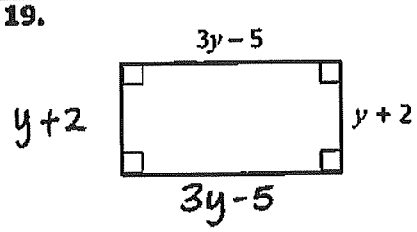
$6x + 1$

18.



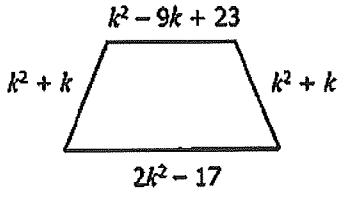
$4a + 4ab + 16$

19.



$8y - 6$

20.



$5k^2 - 7k + 6$

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